

Full paper

Piezoelectric diphenylalanine peptide for greatly improved flexible nanogenerators

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ABSTRACT

We explored the piezoelectric behavior of diphenylalanine (FF) peptide using multiphysics finite element models. We compared the piezoelectric potential developed in a finite element model of an FF peptide nanowire with the potential developed in conventional piezoelectric materials. The simulation showed that the FF peptide nanowire can generate significantly higher voltage than nanowires made from piezoelectric materials such as zinc oxide, lead zirconate titanate, and barium titanate under the same force. The simulation work was expanded to create a complete model of a flexible nanogenerator based on FF peptide microrods. Based on the piezoelectric potentials calculated in the nanogenerator model, we estimated an open circuit voltage for the FF nanogenerator. To validate the model, we fabricated and demonstrated the first flexible piezoelectric nanogenerator based on vertical arrays of FF peptide microrods. We characterized the electrical behavior of the fabricated nanogenerators, and measured an open circuit voltage of up to -0.6 V. The estimated open circuit voltage predicted by the model was in close agreement with the measured value from the fabricated nanogenerator. The results illustrate the promising potential of FF peptide as a piezoelectric material, and show the importance of finite element models for providing insight into the development of a new generation of FF peptide-enabled devices.

1. Introduction

Mechanical energy is present throughout the environment in many forms including wind, water, mechanical vibration, and human movement. The ability to harvest this energy using nanogenerators is enabling the development of self-powered systems such as sensors for structural monitoring, environmental measurement, and chemical and biosensing [1–8]. Piezoelectric materials are especially well suited for harvesting mechanical energy due to their inherent electromechanical coupling. Piezoelectric nanogenerators have been demonstrated using materials such as zinc oxide (ZnO), poly(vinylidene fluoride-trifluoroethylene) (P(VDF-TrFE)) and single layer molybdenum disulfide (MoS₂) [9–18].

Diphenylalanine (FF) peptide is a dipeptide building block that has received much interest due to its biological origins and demonstrated ability to self-assemble. Diphenylalanine peptide is also significant due to its role as the core recognition motif for Alzheimer's beta amyloid, and potential applications for drug delivery [19,20]. Diphenylalanine-based dipeptides have been shown to self-assemble into nanotubes, hydrogels, and other unique structures [21–24]. Self-assembled FF

peptide nanotubes have a hexagonal structure and possess excellent mechanical and piezoelectric properties. Reported values for the elastic modulus range from 19 to 27 GPa, and the origins of this high stiffness have been explored through first principles analysis [25–27]. The piezoelectric properties of FF peptide can be improved by poling the material, with values for the piezoelectric coefficient d_{33} reported to be approximately 18 pm/V [28,29]. Recently, we demonstrated a rigid nanogenerator for mechanical energy harvesting using diphenylalanine (FF) peptide with controlled polarization [30]. The promising electro-mechanical properties, biocompatibility, and low-temperature synthesis method have contributed to the widespread and growing interest in FF peptide as a material for nanoscale applications such as energy harvesting.

While FF peptide shows great promise as a piezoelectric, bio-inspired material, there are still significant barriers that must be overcome to enable its widespread adoption for device applications. The first barrier is the need for comprehensive multiphysics models of piezoelectricity in FF peptide structures. Finite element analysis and first principles have been used to investigate the mechanical properties of FF peptide microstructures and bulk material [25,27]. Additional

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work is needed to investigate the piezoelectric performance of FF peptide structures in comparison with conventional piezoelectric materials. Additionally, current models do not explore the complex interactions and piezoelectric behavior of the FF material as part of a completed device. The fabrication and testing of new devices is high-risk, complex, and time consuming. Without comprehensive models, there is insufficient information to fully predict the response of FF peptide devices, which poses a challenge for device design. Finite element models have previously been used to investigate the behavior of piezoelectric nanogenerators based on materials such as ZnO [31–33]. We present a multiphysics finite element model of an FF peptide nanogenerator with the hope of providing insight into the design, optimization, and research of FF peptide-based devices.

The second barrier to FF peptide material adoption is the need to realize the outstanding electromechanical properties of piezoelectric FF peptide in a flexible device. The transition from rigid to flexible substrates is not straightforward, and presents challenges for material growth, characterization, and device fabrication. By overcoming these challenges, we developed and characterized a flexible FF peptide nanogenerator, and demonstrated power generation from bending deformation. We validated the predictions of the finite element model against the fabricated nanogenerator, showing the importance of the finite element approach for predicting device performance. This work can help to expand the use of FF peptide nanogenerators and devices to new applications that are inaccessible to rigid substrates, such as flexible electronics, wearable devices, and soft robotics.

2. Material and methods

2.1. Finite element analysis

Finite element models were created in COMSOL Multiphysics to evaluate the piezoelectric potential of the FF peptide material. The first model was a single nanowire for calculating the potential in a nanowire made of different piezoelectric materials under uniaxial and transverse loading conditions. The second model was a 2D simulation of the peptide nanogenerator. The single nanowire model was validated against previously published results for the potential in a ZnO nanowire. The nanowire used for our comparison study was modeled as a solid cylinder with a diameter of 50 nm and length of 500 nm. The nanowire dimensions were the same for all materials. The bottom surface of the nanowire was fixed and set to ground potential. A force of 10 nN was applied equally across the top surface of the nanowire in two separate cases, as a pure compressive force, and as a pure transverse force. The piezopotential in the nanowire model was calculated for FF peptide and other common piezoelectric materials such as ZnO, PZT, BaTiO₃, and LiNbO₃. Piezoelectric properties for ZnO and other common materials were taken from the library of standard materials in COMSOL Multiphysics. The peptide material properties were taken from measured and derived values reported in the literature. A complete listing of the properties used for the FF peptide simulation can be found in [Supplemental table A.1](#).

The second finite element model was a 2D model of the FF peptide nanogenerator. [Supplemental Fig. A.1a](#) shows a schematic of the FF peptide nanogenerator, and a detailed view of the geometry domains in the COMSOL model. The piezopotential in the microrod array was assumed to be uniform across the width of the device (out of the page direction) due to the uniform loading across the width of the simply-supported beam. This assumption allowed for the use of a 2D model, which greatly reduced the computational time and complexity. A 3D model of the peptide nanogenerator substrate confirmed that the stress distribution was in fact uniform across the width of the device.

The model consisted of a 12 mm long and 0.25 mm tall rectangle representing the PI peptide growth substrate. Peptide microrods were modeled as 7 μ m wide by 50 μ m long rectangles based on measurements taken from the SEM data in [Fig. 4b](#). The microrods were placed in

a row directly on the PI growth substrate with a spacing of 40 μ m. A third rectangle layer sized 12 mm long by 75 μ m thick represented the PDMS layer in which the microrods were embedded. The gold electrode layers were neglected in the simulation due to their relatively small thickness (100 nm). The nanogenerator was attached to a 30 mm long by 0.25 mm tall PET rectangle, which served as an example of a flexible application. Material mechanical properties of PI, PDMS, and PET were taken from the COMSOL material library. PDMS, PET, and PI were modeled as linear elastic materials for this simulation, and linear elasticity was assumed for the model physics. The +y direction was taken as the d₃₃ direction for the piezoelectric material.

The schematic of the nanogenerator model and corresponding geometry domains are shown in [Fig. A.1a](#). The nanogenerator attached to the PET surface was modeled as a simply-supported beam as shown in [Fig. A.1b](#), with a pin at x = 6 mm, and a roller at x = 24 mm. The pin was modeled as a fixed displacement in the x and y directions, and the roller was modeled as a fixed displacement in the y direction. A uniform pressure of 32 kPa was applied along a 3 mm section of the nanogenerator as shown in [Fig. A.1b](#). The 3 mm length was the approximate contact length of the push rod used for mechanical testing, and the applied load created an equivalent deflection of 3.5 mm at the center of the device. The equivalent distributed load was used instead of directly applying a prescribed displacement because the distributed load created a more realistic deformation in the contact area of the model. A sample of the mesh showing the microrods can be seen in [Fig. A.1c](#).

2.2. FF peptide growth and nanogenerator fabrication

Polyimide film squares (DuPont Kapton) with dimensions 12 mm $\times$ 12 mm $\times$ 0.25 mm were laser cut to serve as flexible growth substrates for FF peptide microrod arrays. The substrates were sputtered with a 5 nm thick chromium (Cr) adhesion layer followed by a 100 nm thick gold (Au) electrode layer. The growth of FF peptide microrods was discussed in detail in [29], and growth of vertical FF peptide microrod arrays on a flexible substrate is briefly summarized here. A solution of FF peptide (Bachem) powder dissolved in hexafluoroisopropanol (HFP) was dropped onto the PI substrates in a vacuum chamber. The chamber was evacuated and then purged with dry air, resulting in an amorphous FF peptide seed layer. Next, the substrates were placed in a humidity chamber between two parallel aluminum plates. A uniform electric field was applied to the substrates by applying 5 kV to the parallel plates while 100% relative humidity air was circulated across the substrates for 60 s. This resulted in crystallization of the amorphous seed layer. During seed layer crystallization, a non-conductive tape was used to affix the PI substrates between two parallel aluminum plates for poling. In previous studies [29,30], a conductive carbon tape was used to mount silicon substrates between the poling plates. Replacing the conductive tape with the nonconductive tape resulted in a more uniform seed layer on the PI substrates, and the reason for this improvement is worth further investigation. Finally, the crystallized seed layer substrates were floated face down in a 5 ml solution of FF peptide dissolved in water for 6 h at 55 $^{\circ}$ C. During the 6-h growth, the substrates were poled between two parallel aluminum plates under a voltage of 8.5 kV.

The FF peptide microrod arrays were spin-coated with PDMS (Slygaard 184) at 4000 RPM for 2 min, and allowed to cure. Next, a 5 nm layer of Cr and a 100 nm thick layer of Au were sputtered on top of the PDMS to serve as the top electrode. The packaged FF peptide nanogenerators were each affixed to a 30 mm long $\times$ 12 mm wide $\times$ 0.25 mm thick layer of PET for testing.

2.3. Device testing

We fabricated and tested multiple FF peptide nanogenerators. A periodic displacement was applied to the nanogenerator using a linear motor (LINMOT). The device was placed in a 3D printed fixture which

approximated simply supported boundary conditions. A prescribed displacement of 3.5 mm was applied to the center of the device using the linear motor. The displacement was applied uniformly across the width of the device using a rounded push rod end. At maximum deflection, the push rod was in contact with an area 3 mm long across the 12 mm width of the device. The displacement was applied and released periodically, at 1 s intervals. Open circuit voltage and short circuit current were collected using National Instruments data acquisition hardware and LabVIEW, along with a current amplifier and nano voltmeter (Keithley 428, 2182 A). A linear measurement drift was removed from the open circuit voltage data. Voltage and current were measured separately across external load resistors ranging from 200 k Ω to 1000 M Ω to calculate the power output from the nanogenerators.

3. Results and discussion

3.1. Finite element analysis of a single nanowire

A model of a single nanowire was created in COMSOL Multiphysics to explore the piezoelectric behavior of FF peptide as compared to other piezoelectric materials. We validated our nanowire model using geometry and results from a previous study of the piezopotential in a ZnO nanowire [16,34]. After validation, the nanowire model diameter was set to 50 nm, the length was set to 500 nm. The dimensions were held constant for all materials. Electromechanical properties for FF peptide were sourced from various computational and experimental works discussed in the methods section, and provided in the [supplemental materials in Table A.1](#). Piezoelectric properties for all other materials were taken from the COMSOL Multiphysics library of standard materials [35].

Fig. 1 compares the piezoelectric potentials developed in a single FF peptide nanowire and a single ZnO nanowire. **Fig. 1a** shows the potentials under a 10 nN compressive load, and **Fig. 1b** shows the potentials under a 10 nN transverse load. In all cases, the nanowire was fixed and grounded at the bottom surface, and the load was applied uniformly to the top surface. The potential developed in the peptide nanowire was about five times higher, developing -1.3 V under compression, compared with -0.26 V for ZnO. The transverse loading voltages were ± 0.24 V and ± 0.039 V for FF peptide and ZnO, respectively. The $\pm$ indicator for potential under transverse loading is due to the opposite potentials developed on the tension and compression sides of the nanowire. Previous models of the piezoelectric potential in a bent ZnO nanowire showed a region of reversed potential at the base of the nanowire [16]. This characteristic potential reversal was also observed in our finite element model of a bent ZnO nanowire as shown in **Fig. 1b** (right), further validating the model. Interestingly, this

potential reversal was also seen at the base of a bent nanowire made of FF peptide, which can be seen in **Fig. 1b** (left). The insets of **Fig. 1b** show a detailed view of the potential reversal at the root of the nanowire.

We further investigated the scaling effects in the single nanowire model. The diameter and length of the nanowire were scaled together by a factor 2, 10, and 100. The largest nanowire model had dimensions comparable to that of the microrods used in the 2D flexible nanogenerator model. The electric potential was calculated in the scaled nanowire models of FF peptide and ZnO for both compressive and transverse loads. The ratio between the potential in the FF peptide nanowire and the potential in the ZnO nanowire remained constant as the nanowire dimensions were scaled. This result suggests that the FF peptide material properties are scalable for both compressive and transverse loading conditions.

We investigated the piezoelectric potential created in the nanowire model by conventional piezoelectric ceramics, including lead zirconate titanate (PZT), barium titanate (BaTiO₃), and lithium niobate (LiNbO₃), and compared the simulation results with the piezoelectric potential created by FF peptide and ZnO. When the same force was applied to a nanowire of the same dimension, the simulation results showed that FF peptide generated significantly higher potential than any other materials, in both normal and transverse modes. A summary of the results for piezoelectric potentials are presented in the [Supplemental in table A.2](#).

The higher piezoelectric potential developed in the simulated FF peptide nanowire is attributed to a combination of electrical and mechanical properties. The relative permittivity of FF peptide ($\epsilon_{33} = 4$) is lower than that of ZnO ($\epsilon_{33} = 12$), while the d_{33} value (18 pm/V) exceeds that of ZnO (11.7 pm/V). Compared to PZT-2 ($\epsilon_{33} = 450$), the relative permittivity of FF peptide is lower by 2 orders of magnitude, while the d_{33} value for FF is only lower than that of PZT (152 pm/V) by one order of magnitude. Additionally, although FF peptide is remarkably stiff for a bio-inspired material ($c_{33} = 24$ GPa), it is relatively compliant in comparison to conventional piezoelectric ceramics such as ZnO ($c_{33} = 211$ GPa) and PZT-2 ($c_{33} = 113$ GPa). Different from PZT and ZnO, d_{15} and d_{24} are significantly greater than d_{33} in FF peptide. As such, shear strains can make a significant contribution to energy conversion with FF peptide. Therefore, for a given force, the FF peptide material will experience higher strain, contributing to the higher piezoelectric potential. Simulation of energy conversion of FF peptide under different strain conditions is worth further investigation.

3.2. Finite element analysis of FF peptide flexible nanogenerators

We expanded our simulation work to further explore the piezoelectric behavior of FF peptide microrods in a device application. We created a 2D model of a complete FF peptide-based flexible nanogenerator structure using COMSOL Multiphysics. The nanogenerator model consisted of a flexible polyimide (PI) substrate with an array of vertical FF peptide microrods. The microrods were modeled as rectangles with a width of 7 μ m and a length of 50 μ m. The microrod dimensions were estimated from SEM images of actual synthesized FF peptide microrods. The microrods were encapsulated in a polydimethylsiloxane (PDMS) layer, and the nanogenerator was attached to a thin piece of flexible polyethylene terephthalate (PET). The flexible PET served to illustrate how the nanogenerator could be attached to a variety of flexible surfaces, for example as a wearable device. We applied a distributed load to the middle portion of the top surface of the nanogenerator that resulted in a 3.5 mm displacement at the center. A schematic of the nanogenerator model with boundary conditions is provided in the [Supplemental material in Fig. A1](#). **Fig. 2a** shows the initial state of the nanogenerator, and the bending stresses in the deformed state. Discontinuities in the stress distribution can be seen due to the different elastic moduli in the PET, PI, and PDMS layers as shown in **Fig. 2b**. The maximum and minimum stresses in the beam were 27.3 MPa and -92.2 MPa, respectively. The strain, shown in **Fig. 2c**, was continuous

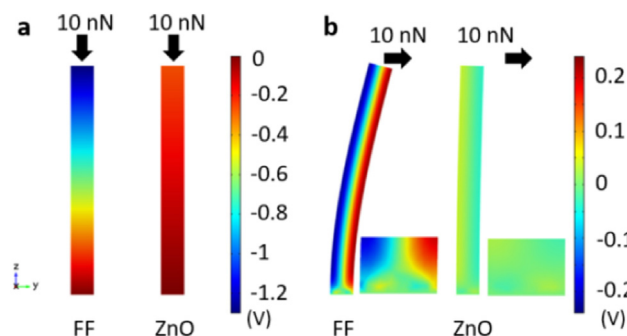


Fig. 1. Simulation of electric potential in 500 nm long by 50 nm diameter nanowires. The bottom surface of the nanowire was fixed and set to ground. The load was applied evenly across the top surface. (a) Piezoelectric potential in a y-z plane slice for FF peptide (left) and ZnO (right) nanowires under 10 nN compressive force. (b) Piezoelectric potential in a y-z plane slice for FF peptide (left) and ZnO (right) nanowires under 10 nN transverse force. Insets show regions of potential reversal at the nanowire roots. Deformations are to scale.

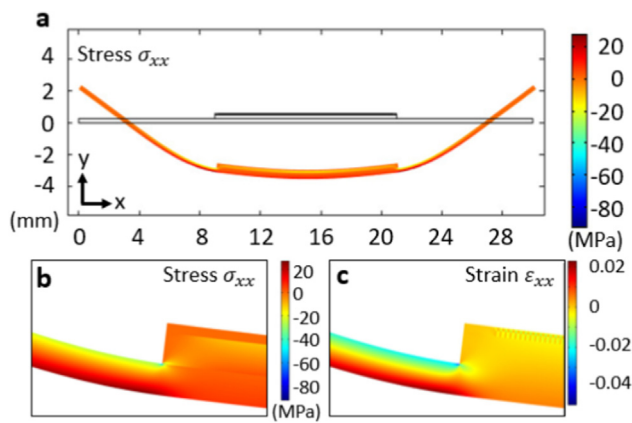


Fig. 2. Stress and strain in 2D model of FF peptide nanogenerator. The distributed load was applied in the middle section and produced a maximum deflection of 3.5 mm. (a) Stress distribution in the nanogenerator showing the initial and deformed state. (b) Detailed view of the stress across the material boundaries. (c) Detailed view of the strain across the material boundaries.

between the layers, as expected. With the exception of localized stress concentrators, the strains were generally within $\sim 2\%$, confirming that the device could be modeled using linear elasticity.

Fig. 3 shows the piezoelectric potential in the FF microrods along the length of the nanogenerator under the 3.5 mm deflection. The microrods were grounded on the bottom surface. The electric potential was sampled at the top of each microrod along the length of the device and plotted in Fig. 3a. The vertical axis shows the electric potential, and the horizontal axis is the position along the nanogenerator substrate. The potential was sampled at a point in the center of the top surface of each microrod. In reality, the simulated potential at the top surface of the microrod was slightly non-uniform, due to the complex strain state. The actual variation in potential across the top surface of each microrod was approximately 0.1 V or less.

As shown in Fig. 3a, the electric potential can be divided into three regions: Region II in the middle of the nanogenerator where the distributed load was applied, and Region I and III towards the outside edges of the nanogenerator where no distributed loads were applied. Region I and Region III showed similar electric potentials due to the symmetry of the device. The piezoelectric potential in Region I and Region III decreases slowly towards the center of the nanogenerator, from about -0.2 V to about -0.3 V. At the interfaces with Region II, the piezoelectric potential drops rapidly to -1.6 V. Fig. 3b shows the potential in a group of microrods in Region I, at $x = 11$ mm. Fig. 3c

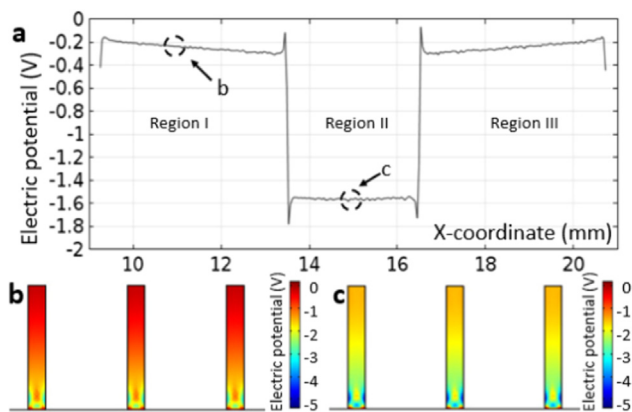


Fig. 3. Electric potential in a 2D model of the FF peptide nanogenerator. (a) Electric potential at the top of the FF microrods as a function of position on the deformed nanogenerator. Electric potential in a group of microrods from (b) Region I and (c) Region II.

shows the potential in a group of microrods in Region II, at $x = 15$ mm. The electric potential within the microrods shown in Fig. 3b,c changes gradually along the length of the microrods. The exception is near the root of the microrods, where a complex potential distribution exists. The complex stress and potential in the root area is likely due to the interface between three different materials, the FF microrods, the PI substrate, and the PDMS encapsulation layer. A more detailed view of this interface location is provided in the Supplemental material, in Fig. A.2 and Fig. A.3.

We investigated the effects of the nanogenerator model geometry on the piezoelectric potential distribution in the microrods. First, we varied the thickness of the PDMS encapsulation layer around the microrods. The PDMS layer thickness was incrementally increased or decreased from the original $75 \mu\text{m}$ thickness. Supplemental Fig. A.4 shows the piezoelectric potential sampled at the top of each microrod along the length of the nanogenerator for different PDMS thicknesses. Increasing the PDMS thickness did not have an appreciable effect on the piezoelectric potential. The piezoelectric potential distribution in the microrods across the nanogenerator surface remained nearly identical for increasing PDMS thickness up to nearly double the original thickness. Similarly, decreasing the thickness to $70 \mu\text{m}$ also had minimal impact on the potential distribution. As the PDMS thickness was further decreased it approached the $50 \mu\text{m}$ length of the microrods. When the thickness of the PDMS decreased to $60 \mu\text{m}$ and below, the distribution of piezoelectric potential in the nanogenerator shifted downward. Regardless of the change in PDMS thickness, the overall shape of the potential distribution, and the size of the step under the distributed load, remained the same. The maximum displacement of the nanogenerator did not change significantly due to changes in PDMS thickness.

Next, we studied the effect of volume fraction on the potential distribution in the microrods. We incrementally decreased the spacing between the microrods from $40 \mu\text{m}$ to about $20 \mu\text{m}$ in order to increase the total volume fraction of microrods in the model. Supplemental Fig. A.5 shows the sampled electric potential at the top center of each microrod, as a function of position on the nanogenerator. As the volume fraction increased, the electric potential distribution shifted upward. The overall shape of the potential distribution, and the size of the step under the distributed load, remained the same. For a given distributed load, the maximum deflection of the nanogenerator was virtually unchanged for different microrod volume fractions. This suggests that increasing the volume fraction did not affect the flexibility of the nanogenerator in the simulation.

Lastly, we investigated the behavior of the 2D flexible nanogenerator model using different piezoelectric materials for the microrods. The nanogenerator was modeled using FF, PZT, and ZnO microrods, while the geometry and applied load were held constant. Supplemental Fig. A.6 shows the electric potential along the nanogenerator surface. As in previous cases, the potential was sampled at a point at the center of the top of each microrod. A jump in potential can be seen in region II for each piezoelectric material, however the jump is largest for the FF peptide material. To explain the difference in the size of the potential jump, we investigated the strain in the microrods. The yy component of the strain tensor was sampled at the same points where the electric potential was measured in the microrods. For all materials, there was a step in the yy component of the strain in region II corresponding to the step in piezoelectric potential. The yy strain tensor component was an order of magnitude larger in the FF peptide microrods than in the ZnO and PZT microrods. This result is consistent with the relative stiffness of FF, ZnO, and PZT. The difference in material stiffness and previously discussed differences in relative permittivity and piezoelectric coefficients can explain why the potential jump is higher for FF microrods in region II. The maximum displacement of the deformed nanogenerator was similar for the different microrod materials.

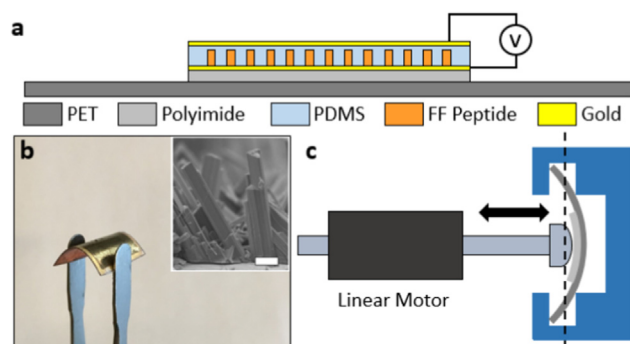


Fig. 4. Structure and testing setup of FF peptide flexible nanogenerator. (a) Schematic of the FF peptide nanogenerator attached to a PET beam. (b) Image of FF peptide nanogenerator highlighting the flexible nature of the polyimide substrate. Inset shows SEM image of peptide microrods. Scale bar is 10 μm . (c) Test setup schematic showing nanogenerator and PET beam with periodic displacement applied by a linear motor.

3.3. Fabrication and characterization of FF peptide flexible nanogenerators

To validate the nanogenerator finite element model, we fabricated flexible FF peptide nanogenerators and compared the measured electrical output with the finite element model results. Fig. 4a shows a schematic of a fabricated nanogenerator, which has the same structure as the finite element model in Fig. 2. We grew arrays of vertical FF peptide microrods on flexible, gold-coated PI substrates using a low temperature synthesis method with controlled polarization [29,30], which is briefly described in the methods section. SEM characterization of the peptide growth showed an array of peptide microstructures as seen in the inset of Fig. 4b. The microrods were approximately 50 μm long and 7 μm in diameter. Next, these arrays of microrods were encapsulated in PDMS by spin coating. A second gold layer was sputtered on top of the PDMS to complete the nanogenerator. A picture of the completed FF nanogenerator is shown in Fig. 4b. The FF peptide nanogenerator was attached to a long, thin PET beam. The PET beam represents a typical flexible surface that the nanogenerator could be attached to, such as a wearable device.

We characterized the power generation of the flexible FF peptide nanogenerators under bending deformation. The PET beams with attached nanogenerators were placed in a fixture that approximated the conditions of a simply-supported beam as seen in Fig. 4c. A periodic displacement of 3.5 mm was applied to the center of the nanogenerators. The displacement was applied evenly across the width of the nanogenerators using a linear motor. Fig. 4c shows a schematic of the experimental setup.

When the flexible nanogenerators were under periodic displacement, we achieved an open-circuit voltage up to -0.6 V and a short-circuit current up to 7 nA. The open-circuit voltage of a nanogenerator

shown in Fig. 5a remained at a constant level, which is ascribed to the good dielectric property and negligible leakage current of FF peptide. We conducted long-term measurement of the open-circuit voltage of an FF peptide nanogenerator under periodic displacement, and the peak to peak V_{oc} remained stable over more than 1000 cycles as seen in Supplemental Fig. A.7. The short-circuit current of a nanogenerator is presented in Fig. 5b. Testing was repeated with reversed connections and the outputs were reversed accordingly, confirming the measured signal was truly from the piezoelectric FF peptide instead of the variation of contact resistance or parasitic capacitance. The power output characteristic was measured with external resistors connected to a nanogenerator while it was repeatedly deformed, and the result is reported in Fig. 5c. The maximum power generated by the device was calculated to be 0.1 nW based on a peak current of 1 nA, and a load resistance of 100 M Ω . We compared the open circuit voltage of the flexible FF peptide nanogenerators with flexible nanogenerators made from other piezoelectric materials. The open circuit voltage of our FF peptide nanogenerator exceeded the value achieved by other flexible piezoelectric nanogenerators based on ZnO [36], monolayer MoS₂ [10,37], PVDF [38], and other biological materials such as bacteriophages [39], which is attributed to the high piezoelectric constants and low relative permittivity of FF peptide.

The flexible nanogenerator composed of FF microrods and adjacent polymer matrix can be modeled as a capacitor-type device. The capacitance of the device is the sum of the capacitance for the microrods and the capacitance for the PDMS matrix, while the total charges during the energy generation are contributed solely by the microrods due to their piezoelectric effect. The SEM images show that the volume fraction of vertical FF microrods is in the range of 25–33%. If we assume the volume fraction of FF microrods as 30% and take the piezoelectric potential as -1.6 V from the finite element analysis of the FF microrods, we estimated that the nanogenerator would produce a V_{oc} of approximately -0.61 V which was consistent with our experimental measurements. The close agreement between the simulation and measured output confirms the validity of the model. This result illustrates the importance and significance of developing comprehensive finite element models for the design and optimization of FF peptide-based devices.

4. Conclusion

Finite element modeling was used to study the piezoelectric potential in a single FF peptide nanowire, which was found to be higher than that of conventional piezoelectric materials such as ZnO, PZT, BaTiO₃, and LiNbO₃, for a given force. The finite element modeling work was expanded to a complete 2D model of a flexible nanogenerator based on FF peptide microrods. To validate the model, we fabricated the first flexible FF peptide nanogenerator with vertical microrods and successfully harvested mechanical energy from bending deformation.

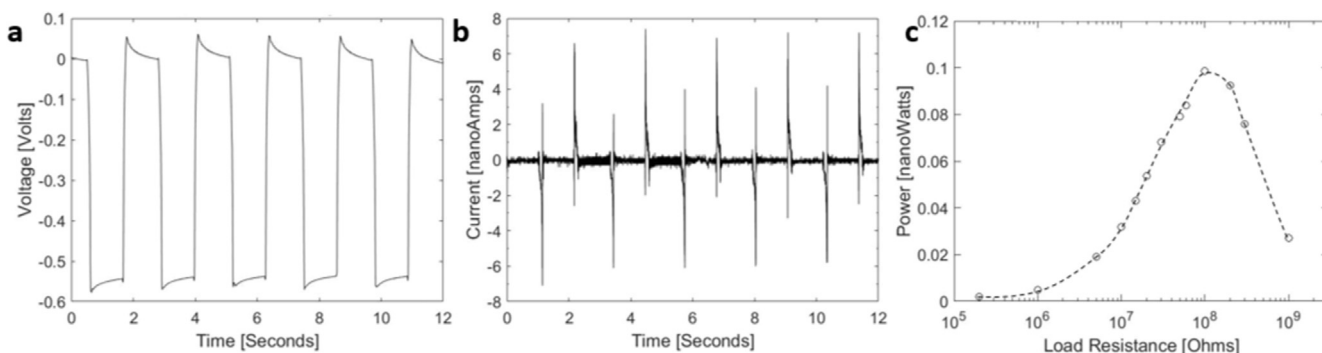


Fig. 5. Characterization of a FF peptide flexible nanogenerator. (a) Open-circuit voltage (V_{oc}) and (b) short-circuit current (I_{sc}) for the peptide nanogenerator under periodic 3.5 mm displacement. (c) Power as a function of external load resistance from 200 k Ω to 1000 M Ω .

We characterized the electrical response of the fabricated nanogenerator and used the results to successfully validate the finite element model of the nanogenerator. The growth of FF peptide on flexible substrates opens the way for new applications such as wearable devices and flexible electronics. Finite element modeling can play an important role in the design of this new generation of flexible FF peptide devices.

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Data availability

The data that support the findings of this study are available on request from the corresponding author (R.Y.)

Competing financial interests

The authors declare no competing financial interests.

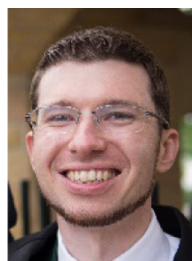
Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.nanoen.2018.06.061>.

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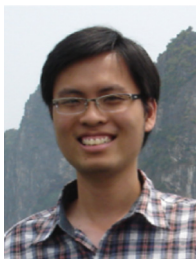
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